

HIGH SECURITY DESIGNS For Remote Controls

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This article discusses the various high security designs for remote controls

Remote-controlled connected devices require security. Simple use cases include remote-operated door or bicycle locks (for smart bicycles used in bike sharing apps). This is usually done with the help of a secure element embedded in the platform. The secure element enables security functions that ensure strong resistance against malicious attacks and facilitate security life management of connected devices.

The secure element is basically a tiny chip that can perform cryptography and keep the secret keys safely hidden from attackers. Unlike microcontrollers (MCUs) and memories, it provides strong protection against attacks like differential power analysis and logical attacks.

Hardware-based security is an effective means of provisioning device keys, and protecting confidentiality and integrity of those keys. With hardware-based security, keys are programmed into security ICs in a secure environment, for example, during chip production or in a secure environ-

ment set up by the manufacturer. Once this is done, security ICs (with configured keys) are shipped to a manufacturer without having to disclose the keys.

Hardware-based security

A spokesperson from Qualcomm Technologies says, "A solid security approach uses a combination of integrated hardware-based security features. All features tightly integrated with the operating system, communication protocols, applications and the cloud help deliver new and exciting Internet of Things (IoT) products and services that also protect security and privacy. All defence technologies having these requirements must be hardwired down to the silicon to power IoT devices.

"At Qualcomm Technologies we have a portfolio of chips and software that spans everything from very high-end application processors to the simplest Bluetooth Low Energy (BLE) sensors. We design hardware and hardware-protected firmware to provide security features and superior robustness compared to software-only security solutions."

Sambit Sengupta, associate director - field applications, Avnet India, says "The process of integrating a secure element in the industrial IoT ecosystem or device platform can become complex. The secure element can be used to provide security as a companion chip to most MCUs. We at Avnet help program the secure element securely and personalise it with unique IDs and secret keys. Multiple end users can benefit from this, where SIM cards, banking cards and so on are used.

"For commercial end applications, we find that brand protection and fake/spurious material authentication are required. We provide security solutions customised for such end uses. For example, one large water filter manufacturer wanted to protect its switched-mode power supply from fake brands. Such solutions were deployed to help him."

Fig. 1: Architecture of high security designs for remote control (Credit: Infineon Technologies)

